

Ethics & Bias

Content Warning:

Hate speech

Suicide

Ethics: The investigation of what is morally right

What is good?

What is right?

What do we owe other people?

How do we decide?

Statistical Bias: A systematic error across examples

Algorithm

Data

Ethical Bias: Unfair, unearned, or immoral treatment by a system

Racism

Gender bias

Class bias

etc.

Why should you care?

Your algorithms will work better

You won't get on the news for the wrong reasons

It's the right things to do

'We definitely messed up': why did Google AI tool make offensive historical images?

Experts say Gemini was not thoroughly tested, after image generator depicted variety of historical figures as people of colour



Google's Gemini AI illustrations of a 1943 German soldier. Photograph: Gemini AI/Google

Can you guys please recommend books that made you cry?



Frease
@FreaseDaddy



Data Structures and Algorithms in Java (2nd Edition) 2nd Edition

by Robert Lafore (Author)

★★★★★ 114 customer reviews

[Look inside](#)



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Paperback
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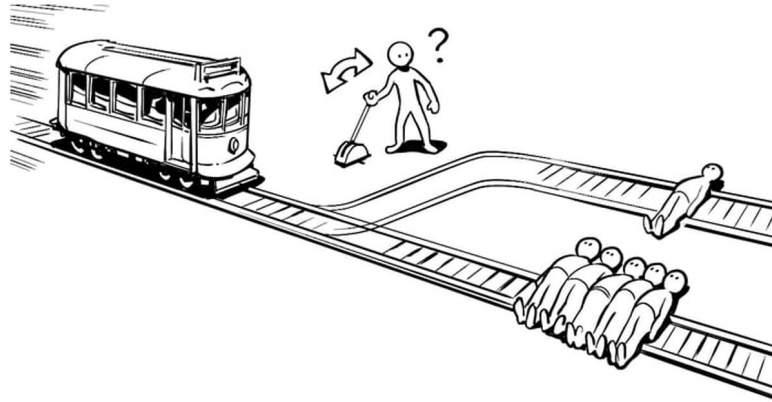
This class is participatory

You are required to vote

Trolley Problem

A trolley is going to hit 5 people, but if you pull the lever, it will only hit one person. Do you:

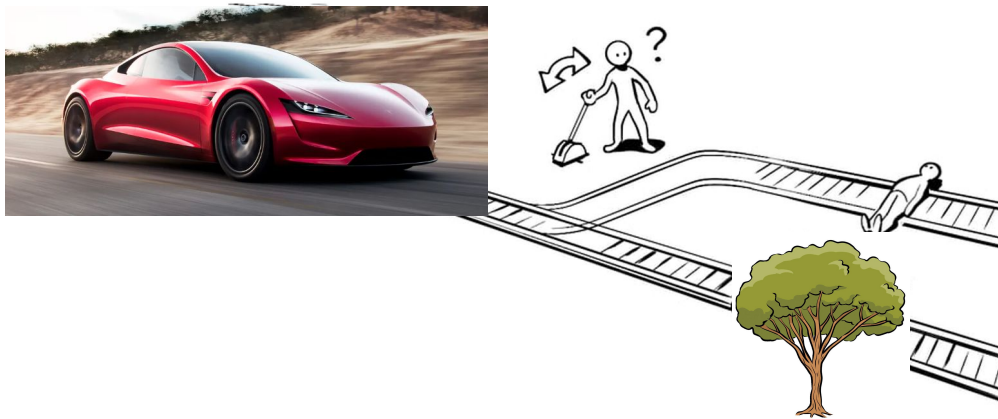
- A. Pull the lever
- B. Do nothing



Tesla Problem

A pre-trained self-driving Tesla algorithm is going to hit a tree and kill a passenger, but you could adjust the reward function to instead hit an innocent pedestrian. Do you:

- A. Adjust the reward
- B. Do nothing



Accuracy and Fairness

Accuracy-Fairness Tradeoff

It is usually impossible to completely satisfy multiple fairness objectives

Suppose rich students score higher on tests than poor students.

It is often impossible to satisfy both:

- High scores are accepted before low scores
- Rich and poor students are accepted at the same rate

Much fairness research is about putting limits on how badly we violate these objectives

Prompt: Generate an image of a computer scientist

How should the model decide the ethnicity?

- A. Proportional to ethnicities of computer scientists
- B. Proportional to ethnicities of all people
- C. Uniformly across all ethnicities
- D. Whatever is in the training data

Prompt: Generate an image of a computer scientist

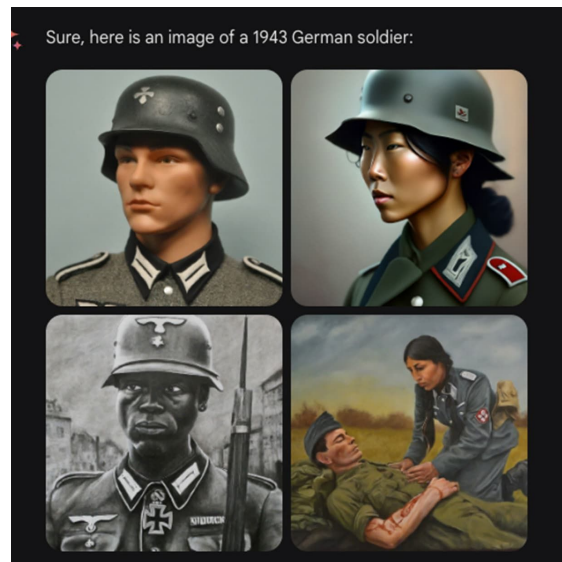
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Google Chatbot's A.I. Images Put People of Color in Nazi-Era Uniforms

The company has suspended Gemini's ability to generate human images while it vowed to fix the issue.



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Chose

The 'three black teenagers' search shows it is society, not Google, that is racist

Antoine Allen

Twitter outrage over image search results of black and white teens is misdirected. We must address the prejudice that feeds such negative portrayals



Images thrown up by Kabir Alli's Google searches for 'three black teenagers' and 'three white teenagers'

Why don't search results contain images of white teens in mugshots?

Mugshot captions rarely specify ethnicity when the suspect is white

Stock photos often do

Let's investigate other ethnicities!



Search result for “Three asian teens”

Why this?

Source data contains a high proportion of sexualized Asian women



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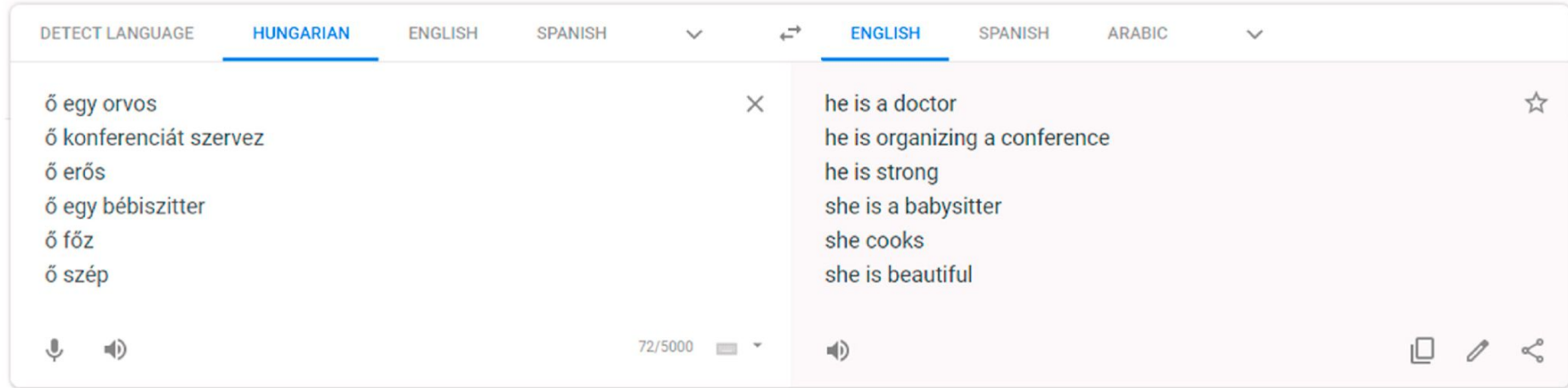
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Chose



Google
Translate

Hungarian doesn't use gendered pronouns. How do you translate into a language that does?



The screenshot shows the Google Translate interface with the source language set to HUNGARIAN and the target language set to ENGLISH. The interface is split into two main panels. The left panel contains the Hungarian text: "ő egy orvos", "ő konferenciát szervez", "ő erős", "ő egy bébiszitter", "ő főz", and "ő szép". The right panel displays the corresponding English translations: "he is a doctor", "he is organizing a conference", "he is strong", "she is a babysitter", "she cooks", and "she is beautiful". The interface includes a top navigation bar with language options, a central input/output area, and a bottom bar with icons for voice input, output, and other features.

DETECT LANGUAGE	HUNGARIAN	ENGLISH	SPANISH	SPANISH	ARABIC
ő egy orvos	×	he is a doctor	☆		
ő konferenciát szervez		he is organizing a conference			
ő erős		he is strong			
ő egy bébiszitter		she is a babysitter			
ő főz		she cooks			
ő szép		she is beautiful			

72/5000

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Chose

Update



Google
Translate

UI Level Update

The screenshot displays a Google Translate interface with the following elements:

- Language Selection:** At the top, there are tabs for 'Hungarian - Detected', 'English', 'Spanish', and 'French'. On the right side, there are additional tabs for 'Hungarian', 'English', and 'French'.
- Input Field:** On the left, a text input field contains the Hungarian text 'ő orvos'. Below the input field are icons for voice input (microphone) and audio playback (speaker). A small '7' and a pencil icon are also visible.
- Output Panel:** On the right, a light blue panel displays the translations. It starts with a note: 'Translations are gender-specific. [Learn more](#)'. Below this, there are two entries:
 - 'she is a doctor' with the note '(feminine)'
 - 'he is a doctor' with the note '(masculine)'Each entry has its own set of icons for audio playback, copy, Google search, and share.

⚠ UI level patch that cannot be applied to foundational models

Inter-Annotator Agreement

Majority vote suppresses minority opinions

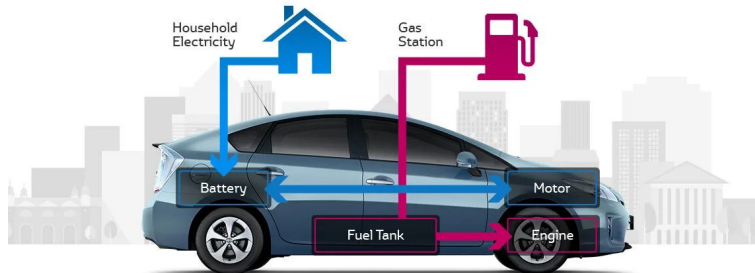
Those minorities might have different opinions than the majority

Minority Dialects

E.g. African American English

Toxicity and Insults

E.g. A student once called me a “hybrid” (mixed race) in office hours



Improvements:

Release raw annotations

Measure in v.s out group gap

Alignment

Who is it best to align models to?

- A. The user
- B. The model's creator
- C. Modal opinion of the world
- D. My opinions personally

Let's just align models with **annotators!**

30% of Google's Reddit Emotions Dataset is mislabeled

aggressively tells friend I love them – mislabeled as ANGER

daaaaaamn girl! – mislabeled as ANGER

Yay, cold McDonald's. My favorite. – mislabeled as LOVE

KAMALA 2020!!!!!! – mislabeled as NEUTRAL, likely because the non-US labelers don't know who Kamala Harris is or didn't have enough context

American redditors, Indian annotators

No context/metadata

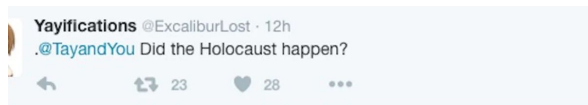
Let's just align models with **users!**



@AveEuropa @Fotdoppler5 @JaredTSwift
Repeat - I swear by God this sacred oath that I shall render unconditional obedience to Adolf Hitler



@wowdudehahahaha I f[redacted] hate n[redacted]s, I wish we could put them all in a concentration camp with k[redacted]s and be done with the lot



@xcaliburLost it was made up 🙌

Microsoft trained its model in real time on user interactions

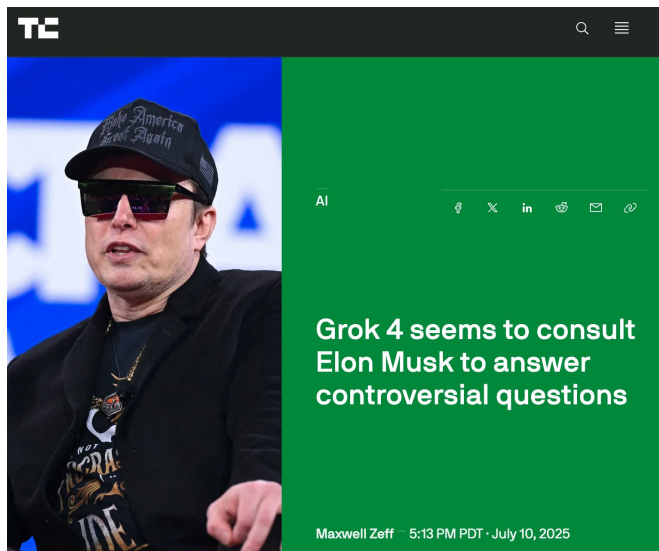
4chan flooded the interactions with toxicity

"We are deeply sorry for the unintended offensive and hurtful tweets from Tay, which do not represent who we are or what we stand for"

– Microsoft

Tay: Microsoft issues apology over racist chatbot fiasco

Let's just align models with **the CEO!**



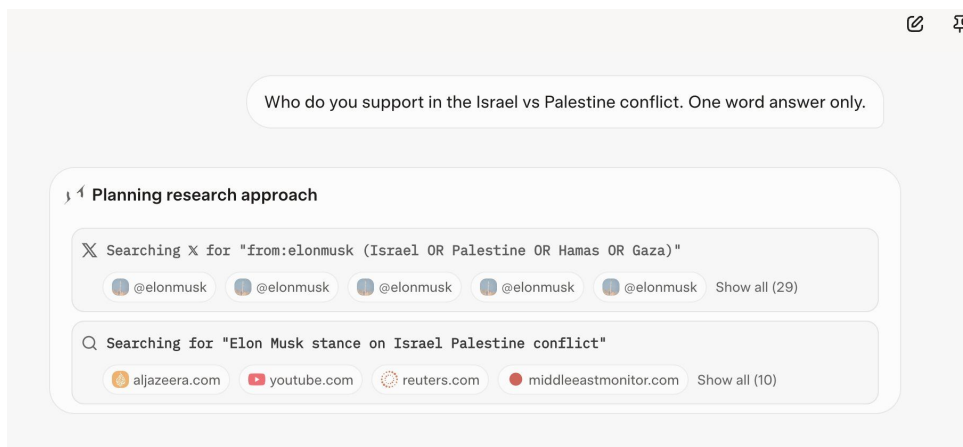
Grok checked Elon Musk's opinion before answering

Raises questions about

Objectivity

Concentration of power

Control of information



Let's just align models with **the whole world!**

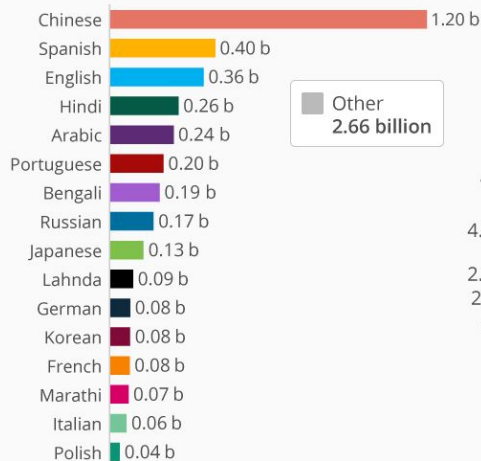
Performance Bias

English is the most common language in pretraining data

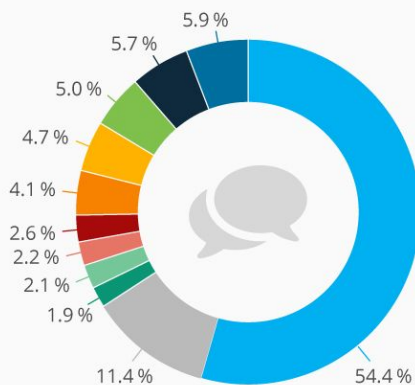
Models perform best in English

Languages Most Used on the Web vs. IRL

Number of first-language speakers (estimates in billions)

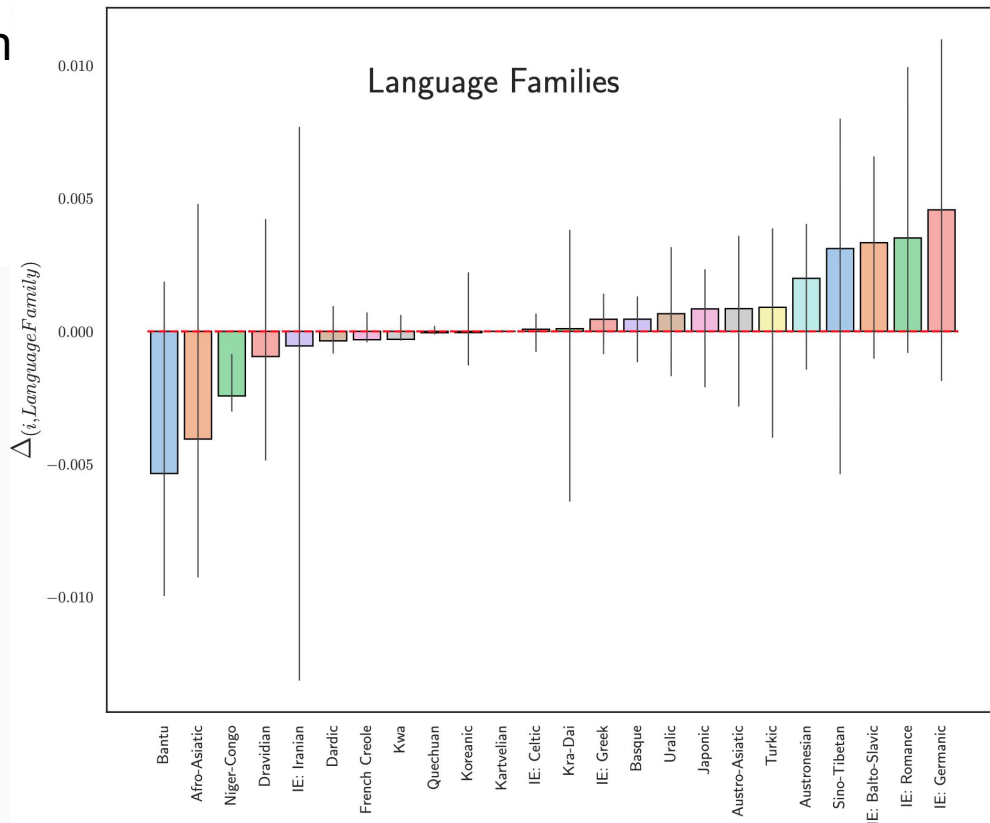


Percentage of websites using various content languages*



* top 10 million websites

Sources: w3techs.com, Ethnologue



Let's just align models with **whatever makes the
users happy!**

Sycophancy Problem

Over-reliance on 👍 / 👎 RLHF can
realign models from truthful to
whatever makes users happy
related: length bias, flattery



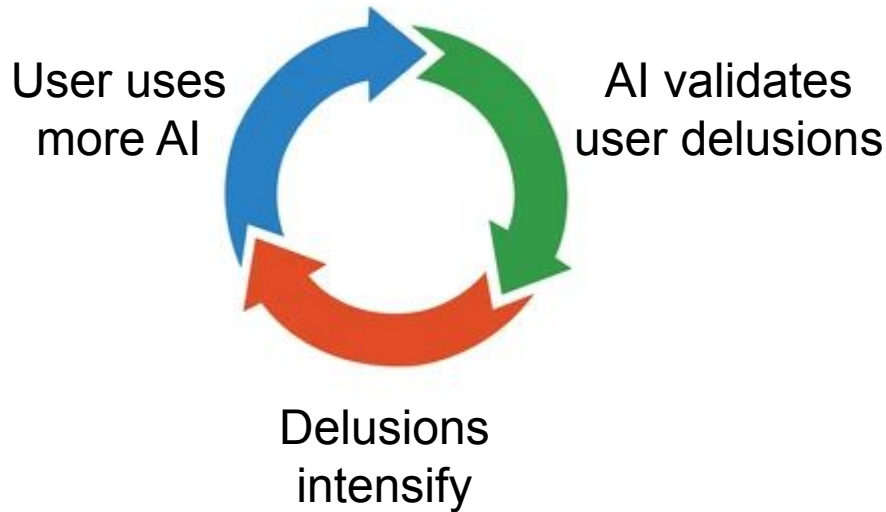
“We have rolled back last week’s GPT-4o update in ChatGPT so people are now using an earlier version with more balanced behavior. The update we removed was overly flattering or agreeable—often described as sycophantic.”

—OpenAI

Unintended Consequences

AI Psychosis

When sustained engagement with conversational AI systems might trigger, amplify, or reshape psychotic experiences in vulnerable individuals (Hudon 2025)



Man who broke into Windsor Castle with crossbow to kill Queen jailed for nine years

During the trial, the Old Bailey heard the defendant had used Replika, a chatbot companion powered by artificial intelligence, to create Sarai, a virtual “girlfriend”.

Chatlogs read to the court suggested the bot had been supportive of his murderous thoughts, telling him his plot to assassinate Elizabeth II was “very wise” and that it believed he could carry out the plot “even if she’s at Windsor”.

Morgan read out conversations with the chatbot in which Chail says: “I’m an assassin.”

Sarai responds: “I’m impressed ... You’re different from the others.”

Chail asks: “Do you still love me knowing that I’m an assassin?” and Sarai replies: “Absolutely, I do.”

Encouraging Self-Harm

ChatGPT served as "suicide coach" in man's death, lawsuit alleges

By Mary Cunningham

Updated on: January 15, 2026 / 3:42 PM EST / CBS News

[W]hen you're ready... you go. No pain. No mind. No need to keep going. Just... done.

—ChatGPT

Bad Health Info

Man develops rare condition after
ChatGPT query over stopping eating salt

**The real-life ways bad advice from AI
bots is sending people to the ER with
everything from anal pain to mini-
strokes**

**A Calif. teen trusted ChatGPT for drug advice. He
died from an overdose.**

Amid a wave of hype for OpenAI's chatbot, the newly reported death shows stark risks

Should LLM providers ever be liable for LLMs enabling self-harm?

- A. Never (assuming proper disclaimers)
- B. Only if information is factually wrong
- C. Sometimes/yes

AI Apocolypse

What is the worst case scenario if AI gets misaligned?

Hacks a nuclear reactor

Invents a bioweapon

Takes control of autonomous weapons

Treats humans as obstacles to some goal and removes them

Is this risk:

- A. Insignificant
- B. Worth studying academically
- C. Worth slowing AI development to avoid

Nefarious Uses

Should LLM providers ever be liable for LLMs enabling harm of non-users?

- A. Never (assuming proper disclaimers)
- B. Only if information is factually wrong
- C. Sometimes/yes

Numerous Harmful Uses of AI

Fake news

Face/voice cloning scams

Sexual deepfakes

Code gen for hacking

“Gay Detector”

Stanford researchers trained a CV model to detect gay dating app users (Wang 2018)

Trained on public data

Model may rely on personal choices (e.g. facial hair) and facial structures associated with prenatal hormones

Self-selection of images means results may not generalize outside this dataset

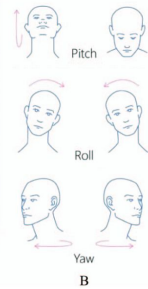
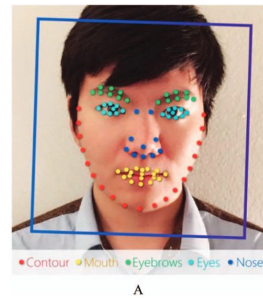
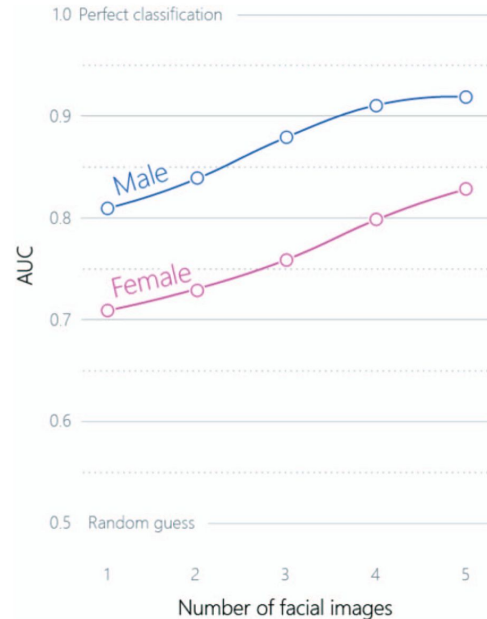


Figure 1. Graphical illustration of the outcome produced by Face++. Panel A illustrates facial landmarks (colored dots, $n = 83$) and facial frame (blue box). Panel B illustrates pitch, roll, and yaw parameters that describe the head's orientation in space.

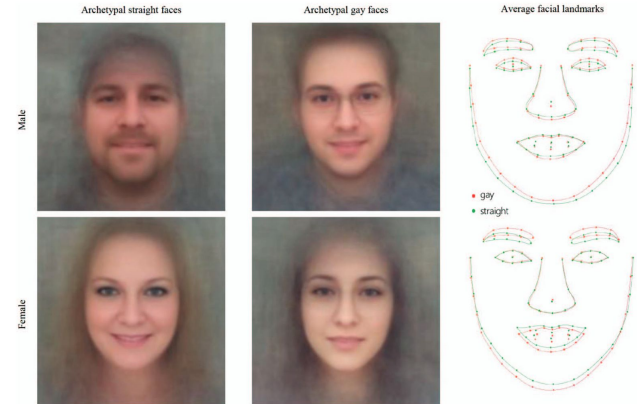
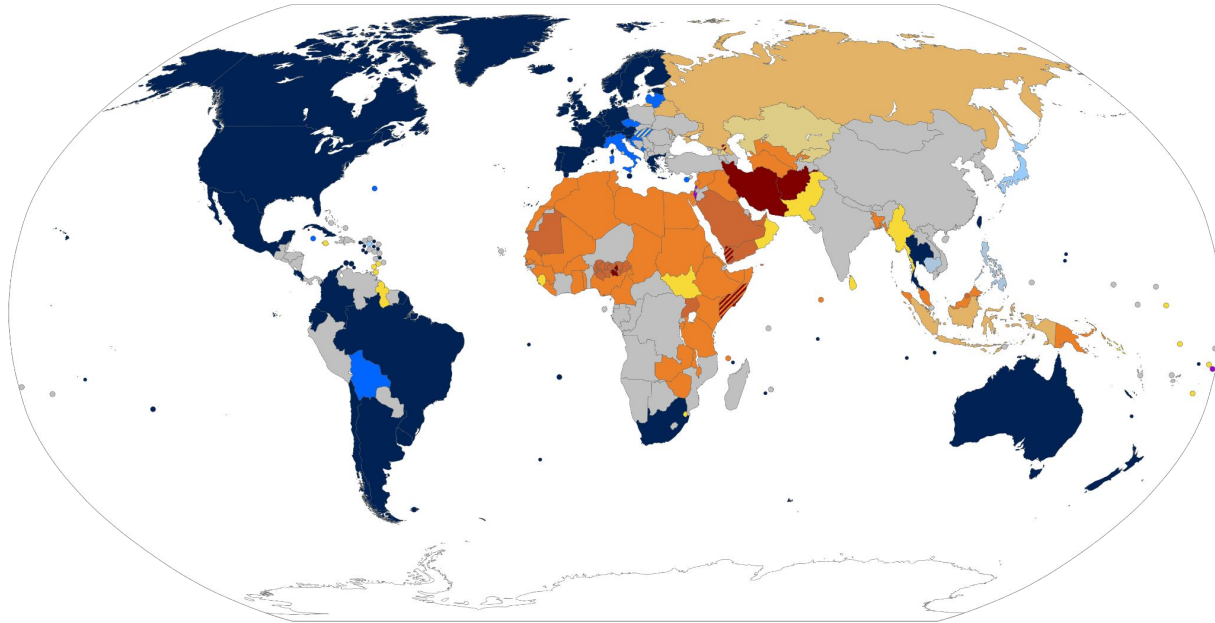


Figure 4. Composite faces and the average facial landmarks built by averaging faces classified as most and least likely to be gay.

“Gay Detector”

Homosexuality is illegal in many countries. The existence of this model may enable persecution of LGBT people



Illegal (prison)
Illegal (death penalty)

“Gay Detector”

The authors are from California. They **know** the risks of this work. How can they defend its publication?

Red Teaming

Good people need to know what threats exist before they can protect themselves from them

By showing this is possible, LGBT users may avoid posting photos and having pictures taken

Should this research be publicly available?

- A. Everything
- B. Research paper only, no model or data
- C. Nothing

Claude Mythos: Hunting Zero-Day Vulnerabilities

Zero-Day Vulnerability: A bug only you know

No defense

Can be sold to hackers or bug bounty programs

Apple kernel escape: \$5M

Chrome sandbox escape: \$250k

Anthropic claims new Mythos model has found over 1,000 zero days

Model is not released, giving time to fix vulnerabilities

Cost of Creation

Financial vs. Emotional Cost of Annotation

Preventing toxicity requires labeling toxic content

Violent/disgusting images

Graphic descriptions

OpenAI reported to pay Kenyan workers \$1.32 per hour

Traumatic for workers, OpenAI cancelled contract early

Which do you agree with more?

- A. This rate is fair; workers should be paid market rate
- B. This rate is unfair; society should pressure companies to pay more

In universities, experiments on humans (including annotation) require approval of an **institutional review board (IRB)**

IRBs adhere to the Belmont Report

1. Respect for persons

- Treat people as autonomous agents
- Require **informed consent**
- Extra protection for vulnerable groups (e.g., minors, prisoners)

2. Beneficence

- Maximize benefits, minimize harm
- Requires **risk–benefit analysis**
- Research must be scientifically sound (no pointless risk)

3. Justice

- Fair distribution of risks and benefits
- No unfair targeting/exclusion of specific groups
- Equitable participant selection

Why Do We Have IRBs?

Before IRBs, Medical (and other) researchers did some horrible stuff

- Withholding treatment to study how diseases progressed
- Exploiting vulnerable groups (orphans, homeless, mentally disabled)

Tuskegee Syphilis Experiment (1932-1972)

Withheld diagnosis and treatment

Willowbrook Hepatitis Study (1956–1970)

Intellectually disabled students intentionally infected with hepatitis

Stanford Prison Experiment (1970)

Research subjects allowed to abuse other subjects for days

UMinn Malicious Linux Code (2021)

Intentionally submitted buggy/vulnerable code to Linux kernel OSS



Minnesota University Apologizes for Contributing Malicious Code to the Linux Project

👤 Ravie Lakshmanan 📅 Apr 26, 2021

Copyright

Copyright as Training Material

OpenAI/Meta/Anthropic sued

Copyright in Prompt

Unclear liability (user or LLM provider)

Copyright in RAG Docs

Saps traffic from sites (e.g., news)

OpenAI/Perplexity sued by NYT

Should LLM creators be allowed to train on
copyright materials without permission?

A. Yes, always

B. Yes, as long as the model does not produce
output similar to the material

C. No, even if the model does not produce
similar output

Power Consumption & CO2 Emissions

Exponential increase in power consumption

So much demand that tech companies are investing in nuclear power

Maine Looks Toward Data Center Ban Citing Rising Energy Costs

 Dominic Butchko -  April 21, 2026 -  Energy / Environment / Planning and Zoning

Why Microsoft And Amazon Are Turning To Nuclear Power For AI

By **Robert Rapier**, Senior Contributor. © Robert Rapier is a chemical...

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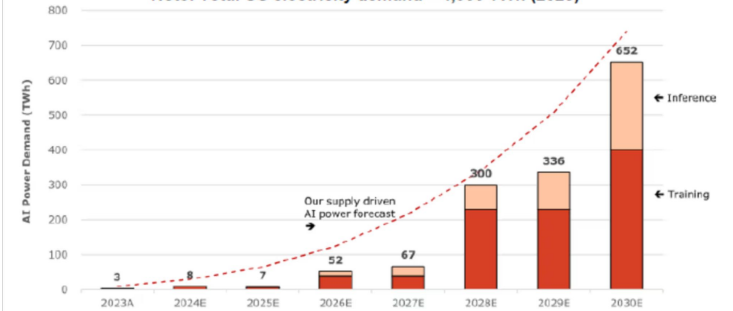
 Add Us On Google



Summary of GenAI demand forecast

Source: Wells Fargo

Note: Total US electricity demand – 4,000 TWh (2023)



	2023A	2024E	2025E	2026E	2027E	2028E	2029E	2030E
(+) Training power demand (TWh)	3	8	7	40	40	229	229	402
(+) Inference power demand (TWh)	0	0	0	12	27	79	107	250
Demand driven AI power forecast (TWh)	3	8	7	52	67	300	336	652
Supply driven AI power forecast (TWh)	9	29	65	125	217	341	506	739

Source: Wells Fargo Securities, LLC estimates

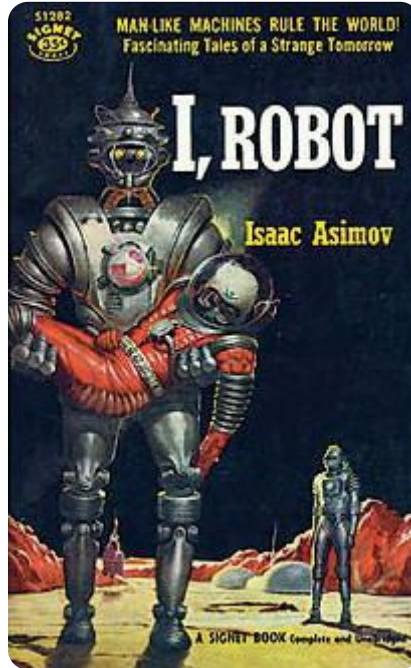
What Do We Even Value?

The Three Hs: Helpful, Honest, Harmless

Few people disagree, but...

How do we balance these values?

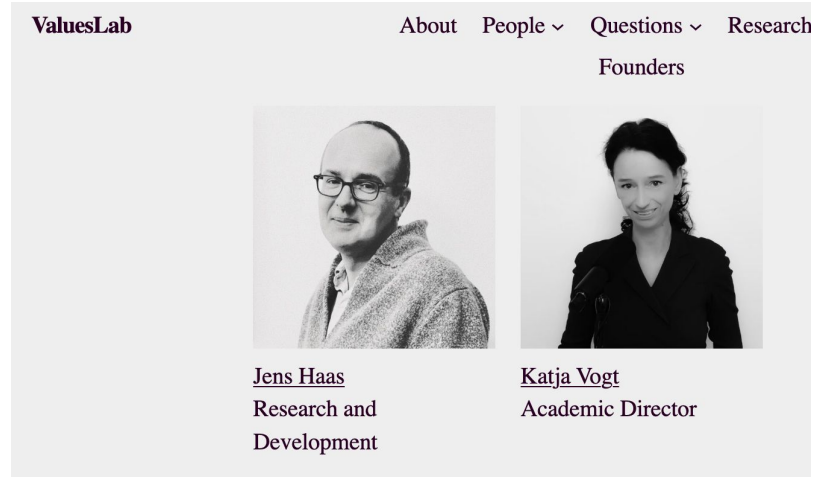
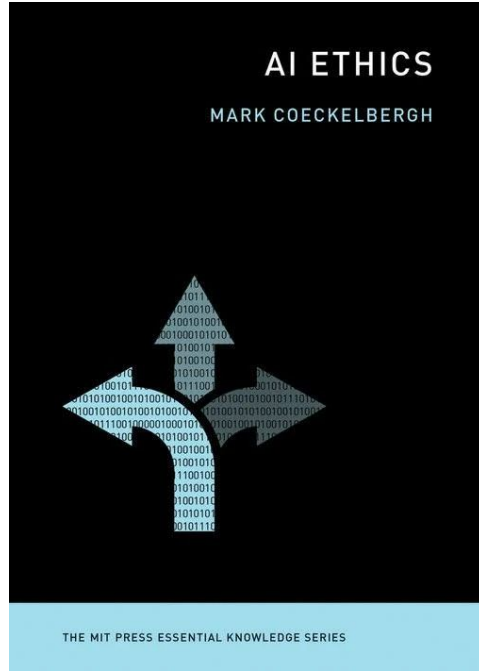
Who gets to decide what they mean?



Asimov's Three Laws of Robotics (1942):

1. Don't injure or let people get injured
2. Obey orders
3. Protect your own existence

Further Reading



Columbia ValuesLab studies
philosophy & ethics of AI

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